

Social Insurance and Occupational Mobility

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Outline

- 1 Introduction
- 2 Methodology
- 3 Results
- 4 Conclusion

Motivation

- Social insurance provides a cushion for workers.
- Differs across countries, affecting labor market dynamics.
- How can we model this effect?

Research Question

Main Question

How does social insurance effect occpational experimentation?

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How does social insurance effect occupational experimentation?

Hypothesis

Providing more social insurance allows for riskier occupational choices.

Model

- Build upon Roy (1951) model of occupational choice.
- Add interaction between earnings risk, social insurance, and occupational choice.

Human Capital

Workers have two types of ability:

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- Innate
 - occupation specific
 - discovered through work experience
- General
 - applicable to all occupations
 - will experience occupation-specific shocks to this human capital

Model Environment

Household:

- Lives for S periods
- endowed one unit of time each period, with no leisure value
- workers dislike risk
- rank levels of consumption c according to a utility function $u(c)$

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Labor Market:

- J occupations, $j = 1, \dots, J$
- workers can only work in one occupation at a time, but can switch between periods
- receive wage w_j per unit of human capital

Value Functions

Value of staying in occupation j :

$$V_s(\Omega_s, z, \epsilon, j) = \{u(c) + \beta \int W_{s+1}(\Omega_{s+1}, z', \epsilon', j') dF(\epsilon')\},$$

s.t.

$$c = T(w_j e^{\theta_j} e^z e^\epsilon)$$

$$z' = z + \epsilon$$

$$\Omega_{s+1} = \Omega_s$$

Value Functions, cont.

Value of switching to occupation j' :

$$H_s(\Omega_s, \theta_{j'}, z, \epsilon, j') = \{u(c) + \beta \int W_{s+1}(\Omega_{s+1}, z', \epsilon', j') dF(\epsilon')\},$$

s.t.

$$c = T(w_{j'} e^z e^{\theta_{j'}} e^{\epsilon'_{j'}} e^{-c(s, \kappa)})$$

$$z' = z + \epsilon'$$

$$\Omega_{s+1} = \{\Omega_s, j', \theta_{j'}\}$$

Equilibrium

$\Psi_{j,s}(\Omega_s, z, \epsilon)$ is the distribution of workers in occupation j at age s with history Ω_s , general human capital z , and shock ϵ . Ψ is defined for all $\Omega_s, z, \epsilon \in \mathbb{R}$

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Mass of newborns:

$$\Psi_{j,0}(\Omega_0, z, \epsilon) = \frac{1}{S} f_j \forall j \in \{1, \dots, J\}$$

where f_j is the fraction of newborns with initial occupation specific ability θ_j .

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For $s = 0, \dots, S$:

$$\Psi_{j,s+1}(\Omega_{s+1}, z', \epsilon) = \sum_{j'} \Psi_{j',s}(\Omega_s, z, \epsilon) l_{j,s}(j', \omega_s, \epsilon, z) \forall j \in \{1, \dots, J\}$$

Equilibrium cont.

Define aggregate mass of efficiency units in occupation j at age s as:

$$N_j = \frac{1}{S} \sum_{s \in S} \int e^z e^{\theta_{j'}} e^{\epsilon_{j'}} + \frac{1}{S} \sum_{s \in S} \sum_{j \neq j'} \int e^{-c(s, \kappa)} d\psi_{j,s}(\Omega_{s-1}, z, \epsilon)$$

Equilibrium cont.

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This let's us define the SCE consisting of

- ① set of occupation level wages $\{w_j\}_{j=1}^J$
- ② occupational populations $\{\psi_j\}_{j=1}^J$
- ③ set of intermediate goods prices $\{p_j\}_{j=1}^J$
- ④ set of occupational labor inputs $\{N_j\}_{j=1}^J$
- ⑤ occupation-specific decision rules $\{I_{j,s}\}_{j=1, s=0}^{J,S}$
- ⑥ value functions $\{V_s\}_{s=0}^S$

Equilibrium Conditions

- ① Above value functions solve optimization problems
- ② Labor inputs N_j are the solution to the intermediate producer optimization problem
- ③ Intermediate goods quantities X_j solve the final goods producer's problem
- ④ Prices p_j equate supply and demand of intermediate goods
- ⑤ Wage in occupation j is the marginal product of an efficiency unit in that occupation s.t.

$$w_j = \alpha_j N_j^{\alpha_j - 1} \prod_{j' \neq j} \{N_j^{\alpha_j'}\}$$

- ⑥ Labor markets clear at occupational level
- ⑦ In occupation j , Ψ_j is the stationary distribution
- ⑧ Final goods market clears by Walras' Law

Data and Calibration

Data Used

- Cross-National Equivalent File (CNEF)
- Panel Study of Income Dynamics (PSID)
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Calibration requires variance of earnings shocks to be parameterized. This is done via estimating various moments

Estimation

- earnings (log): $y_{i,j,t} = \alpha_{ij} +$
- decomposition of risk, $u_{ijt} = \eta_{ijt-1} + \omega_{ijt}$
- ω_{ijt} is permanent and follows a random walk, η_{ijt-1} is a transitory shock which is i.i.d.
- tax function: $y_a = \phi_0 y_p^{1-\phi_1}$
- panel earnings: $y_{it} = \alpha_i +$
- probability of switching: $P_{it} \equiv \Pr(y_{it} = 1 | v_{it-1}^{\bar{\cdot}}) = \psi(v_{it-1}^{\bar{\cdot}}; \beta)$
- mobility cost function: $c(s, \kappa) = \kappa_0 + 1_s + \kappa_{2s}^2$

Main Results: US

- More social insurance leads to a mobility rate of 20%, a 1.3 pp. increase than the baseline
- Workers switch on average 4.3 times in their lifetimes, an 8.2% increase
- We see substitution towards higher risk occupations, leading to a higher average earnings
- Welfare rises by 4.15% under the German scheme via consumption smoothing and better occupational matches

Main Results: Germany

- Less social insurance leads to a decrease in average mobility rate from 2.6% to 2%
- Switching decreases from 0.45 to 0.35 times in a lifetime, due to US tax scheme discouraging occupational experimentation
- Mean earnings decrease in all occupations, most notably in the riskiest ones
- We see workers sort from medium risk to low risk occupations, leading to a polarity of risk types

Conclusion

- Uncertainty in earnings and social insurance are important factors in occupational choice
- Cross-country differences in social insurance can explain differences in occupational mobility and earnings
- Policy implications: designing social insurance programs to balance risk protection and incentives for occupational experimentation